

# Design, Development, Testing, and Flight of the Crew Dragon Docking System

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## Abstract

On March 3, 2019 10:51 UTC, the SpaceX Crew Dragon Demo 1 spacecraft successfully docked to the International Space Station (ISS) on the Node 2 Forward International Docking Adapter (IDA). The SpaceX developed docking system successfully accommodated the misalignments and relative rates between Crew Dragon and ISS, and ultimately attenuated the kinetic energy following contact. This was the first successful docking of a Commercial Crew program vehicle to the International Space Station. The successful Demo 1 flight was the culmination of years of development. This paper describes the SpaceX docking system, and a sample of the lessons learned through simulation, test and inspection.



*Figure 1: The Demo 1 Crew Dragon System on Approach to ISS*

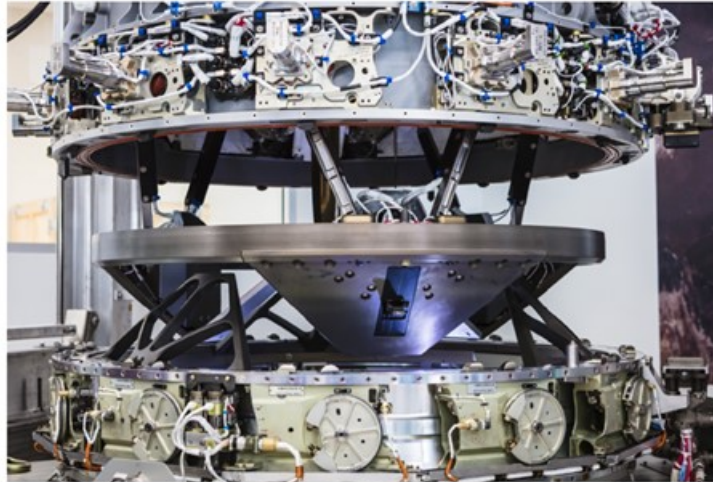
## Introduction

The SpaceX docking system is comprised of several mechanisms. The soft capture system (SCS) allows for Crew Dragon to compensate for misalignments with ISS resulting from navigation and control errors and provides the initial (soft) attachment to the station. The SCS also is responsible for bringing the capsule to rest with respect to ISS, decelerating it from an initial approach speed at contact with ISS of 8 cm/s. The SCS consists of a ring with three coarse alignment petals, each with embedded passive soft capture latches. The SCS petals and passive soft capture latches are derived from the NASA Docking System design.

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*Figure 2: The SpaceX Docking System conducting a joint interface acceptance test; soft capture ring shown in the deployed state*

The SCS is deployed forward of Crew Dragon via a hexapod – a minimally constrained, six-legged arrangement of two-force struts. As the soft capture ring/petal elements make contact with the IDA, forces travel down the rigid struts and are converted into torques by a moment arm connected to a passive rotary spring damper, also referred to as attenuator arm. The rotary spring damper reacts the contact torque with caged helical compression springs and a rotary eddy current damper. The soft capture system passively attenuates the contact loads and allows the soft capture ring to move in six degrees of freedom (6-DoF), permitting it to align with the passive IDA interface. Assuming sufficient alignment is achieved, the soft capture latch pawls jump over corresponding striker plates on the IDA and soft capture is achieved. Two plungers on each soft capture petals are depressed when capture is achieved, triggering a change in microswitch state. If capture is not achieved, Crew Dragon will perform its failed capture response and safely retreat from the ISS.

Following successful soft capture and cessation of relative motion, the soft capture ring is retracted via three linear dual-wound actuators. As the linear actuators are driven up they cause the rotary spring dampers to rotate down and thereby return the soft capture ring to the stowed height. Following soft capture ring retraction, twelve hard capture hooks, derived from the NASA Docking System hook design, connect to corresponding passive hooks on the IDA. The hard capture hooks each contain a cam mechanism that draws the hooks down and compresses a seal that runs around the circumference of the docking system. Ready to hook switches provide the indication that the hooks are in a position to successfully latch onto the IDA passive hooks. Finally, two umbilical dual-wound actuators mate power and data connectors to ISS. Following a successful hard mate operation, the vestibule between Dragon and ISS is pressurized and the hatches opened to allow for transfer of crew and cargo.

To undock, the passive capture latches are moded to a “ready to release” state such that they no longer provide load retention capability. The umbilicals are retracted and the hard capture hooks opened, freeing Dragon from the ISS and allowing GNC to take control to move Dragon away from the ISS. Contingency release mechanisms, Non-Explosive Actuators (NEAs) on the soft capture latches and Frangibolts on the hard capture hooks, are present in the event that release via the nominal actuators fails.

Notably, the hard capture hooks are also utilized to retain the nosecone during launch and re-entry. This protects docking system and other nearby components for reuse.

The SpaceX docking system is a highly performing and efficient spacecraft mechanism. Its weight is approximately 4% of the Dragon 2 mass. It has low power draw and was designed and qualified for five-flight reusability with minimal maintenance.

The SpaceX docking system is a well understood mechanism, which has been computer modeled, simulated, and test correlated to a high degree. The system has been modeled in NASA's Trick software framework, so that this simulation can easily interface with other NASA docking simulations. At many levels of individual components (springs/dampers), sub-systems (soft capture latches, hard capture hooks, rotary attenuator arms), and full docking systems, the simulation has been correlated to match test data from both SpaceX and NASA testing. The correlation of the simulation extends to a wide variety of mass properties in the Crew Dragon class, at a wide range of contact conditions and velocities.

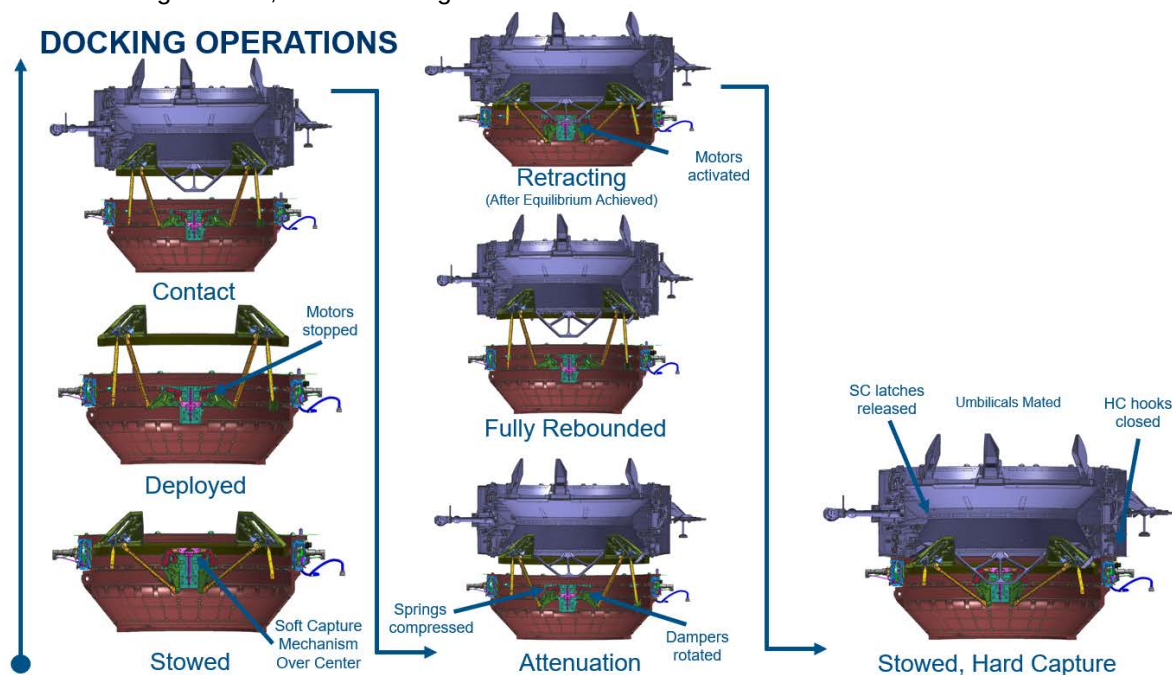


Figure 3: The SpaceX Docking System Operations Overview

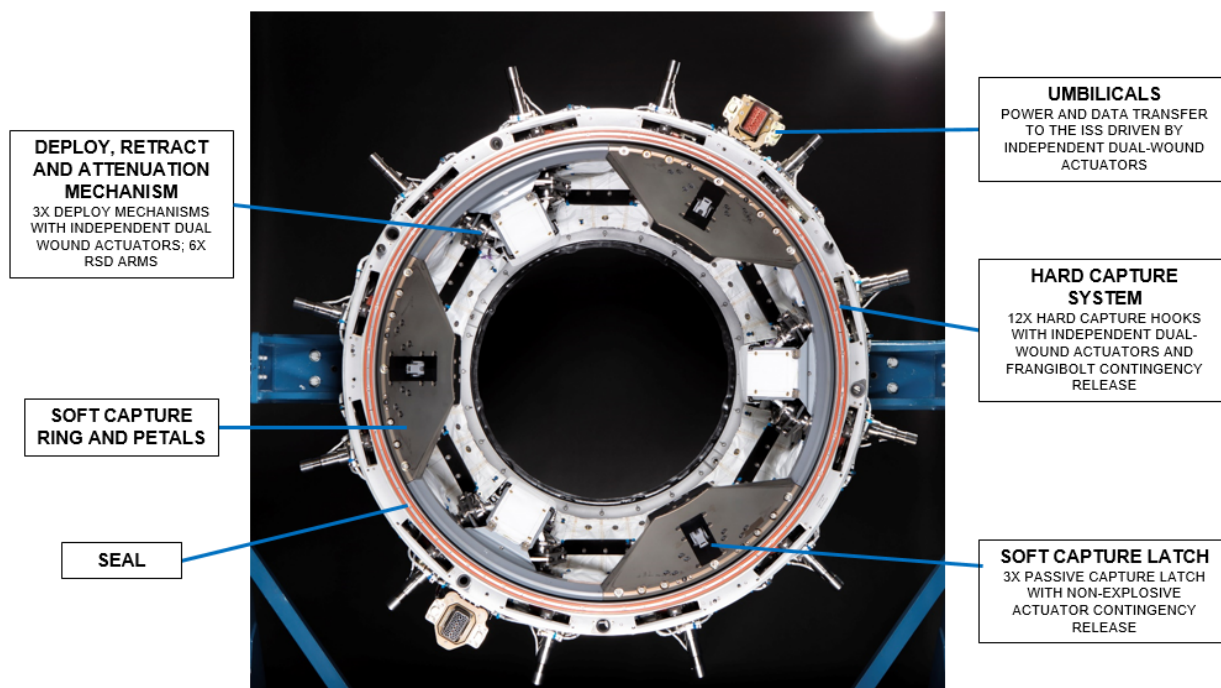


Figure 4: SpaceX Docking System Subassembly Overview

## **SpaceX Development Approach**

The SpaceX team has taken a very integrated, systems-based approach to the development of the SpaceX docking system. This integrated approach has been a cornerstone through early concept development, qualification, and flight integration, build and test.

Starting with the early concept development, close interaction between GNC and the design engineering teams ensured accurate system requirements for misalignments and speeds and informed design selection. During the prototype phase, the mechanisms, software and avionics teams worked tightly together to identify the optimum systems design.

A system-level qualification campaign was undertaken as early as possible to inform docking performance parameters. The SpaceX docking system has undergone extensive qualification testing at both the system and subsystem level, utilizing SMC-S-016 and NASA-5017 mechanisms requirements. The system as a whole completed shock testing, vibration testing, thermal-vacuum and static structural testing. These system level tests included full system performance tests at thermal extremes. This system level qualification testing highlighted interactions that may have been missed with solely subassembly qualifications, such as dynamic harnessing effects. This testing campaign allowed the validation and refinement of flight software throughout the testing process.

In addition, the soft capture system was subjected to extensive performance testing on the Six-Degree-of-Freedom Dynamic Test System (SDTS) at the NASA Johnson Space Center (JSC). The SDTS was used to simulate the relative dynamics of Crew Dragon docking to ISS and was critical to understanding system level performance and interactions. Through two testing campaigns, the SpaceX docking system completed more than 450 soft capture events. Significantly, the system and subassembly qualification testing allowed validation of the docking dynamics simulation.

Throughout the first and now subsequent flight builds, the close interaction between the integrated teams at SpaceX, as well as dynamics modeling team, has been maintained to ensure each docking system is built and validated to system requirements.

### **Sub-System Modeling, Correlation, and Lessons Learned**

The SpaceX Docking System is modeled in NASA/JSC's Trick environment. This docking dynamics model was developed by Dynamic Concepts Inc. as a contractor of SpaceX, and highly tuned via subassembly and system level testing. Simulations were created for docking events as well as sub-scale testbeds, which allows for highly tuned as-built data. This simulation was then able to correlate to Demo 1 docking.

#### Sub-System Lessons Learned – Attenuator Arm Acceptance Testing

In general, while the attenuator arm is conceptually simple with two springs and a damper, the complexity of modeling comes from non-linear effects. For the modeling of initial response and peak loads and dynamics, most components can be modeled linearly (all mechanisms have some degree of non-linearity). However, effects like dead bands, frictions, gaps, local stiffnesses, all have a major role in accurately simulating the behavior of the attenuator arms.

In particular, the correlation of the attenuator arm simulation resulted in several findings regarding the carriage and the damper. Initial modeling of the deploy structure, or carriage, assumed it was static device, but through correlation with ATP test data and additional testing specific to the carriage, it was found that the carriage not only has significant dead band but also has important back structure stiffness. These effects are critical to model properly due to the high stiffness of the rebound spring. The nature of the eddy current damper, with its high gearing reduction results is a slight dead band when velocity reverses. This is a minor effect but becomes more significant at the end of attenuation when the relative velocities between the Dragon and the ISS are small. An example of the correlation in the forces in the strut connecting linkage between the acceptance test and simulation is shown in Figure 5.



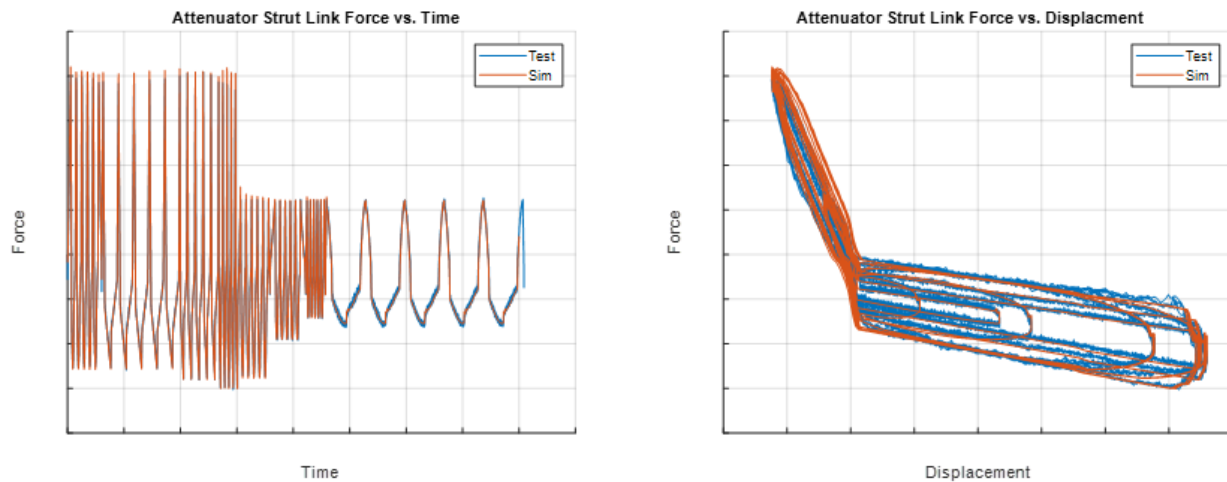


Figure 5: Attenuator Arm ATP Correlation, Test Flight Example Arm Data vs. Simulation

#### Sub-System Lessons Learned – Latch Acceptance Testing

The soft capture passive latch model was derived from a high-fidelity dynamics model, based upon CAD geometry of the latch hardware. The Trick model of the latch did not attempt to simulate every possible degree of freedom but used lookup table data to emulate the nominal behavior of the latch when compressing during docking. The latch model included effects due to the unique geometry of the striker, which is not smooth, and produces force spikes during nominal docking. These effects can be seen below in Figure 6. An important lesson learned was the effect that the significant striker geometry had on latching dynamics, as well as the overall complexity of the latch design and dynamics.

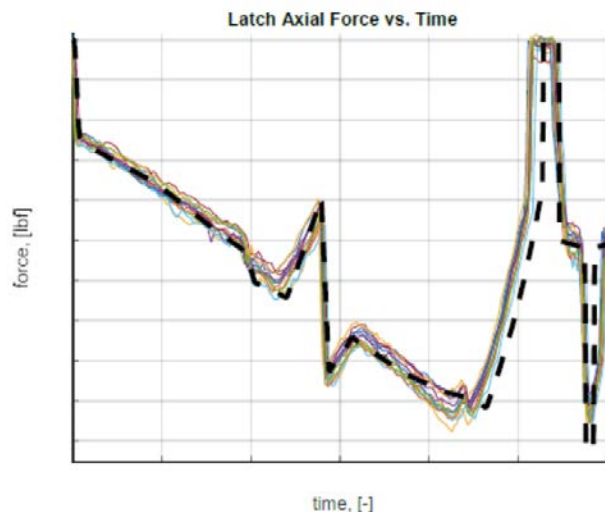


Figure 6: Latch Forces vs. Time, Latch ATP Test data and Simulation data

### **System Modeling, Correlation, and Lessons Learned**

#### System Modeling and Correlation

The Sub-System ATP testing was critical to successfully modeling the SpaceX Docking System behavior when tested at the SDTS facility at JSC. The Six Degree of Freedom Test System facility at NASA's Johnson Space Center enables soft capture, attenuation, and retraction testing of docking systems in a controlled environment.

Correlation of the simulation to SDTS testing could only be accomplished due to integration of the simulation with JSC's simulation of the facility. The facility has several nuances, including controller and actuator effects, which when accounted for, vastly improve the accuracy of the docking simulation. A comparison of the measured forces at the docking plane and the relative vehicle motion, for a representative docking condition, is shown in Figure 7.

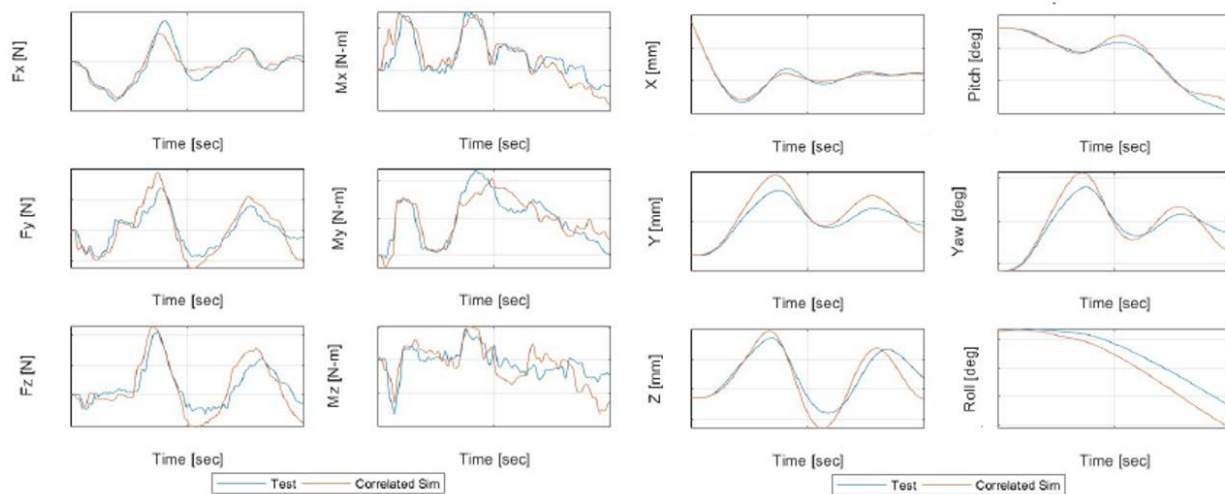


Figure 7: Simulation correlation to SDTS test data, Forces and Displacements vs. Time

Two system test campaigns were performed on the docking system, an initial campaign in 2015 and a second campaign in 2018 with the final flight design iteration implemented. During the first campaign, we learned that the sub-system modeling of the attenuator arms and latches were inadequate. While they could capture the overall behavior of the docking system, there were too many cases which did not correlate properly, many of which were directly due to properly modeling the non-linear behaviors of the attenuator arms. There were also missed captures, which were directly due to improper tolerances of both the latch and striker hardware, but this prompted the creation of latch models of higher fidelity in order to understand the latch behavior.

Several lessons learned came from the model correlation during the second test campaign. Significantly more sub-scale testing was performed on docking system components, which allowed exact spring rates, preloads and damping rates to be modeled and thus leaving only system level variables as unknowns. This higher fidelity modeling of the attenuator arms resulted in much better correlation to a wide verity of test cases, including those that specifically did not correlate as well during the first test campaign. A major lesson learned was that misalignments of the Soft Capture System ring at deployment do not drive docking dynamics. Due to the passive nature of the docking system, and the relatively small pitch and yaw inertias of the Dragon 2, significant misalignments of the soft capture ring at deployment were negligible and did not have a strong impact on capture success. This highlights that there is little benefit to high levels of accuracy in SCS ring position, which can drive complexity for systems with Linear Electro-Mechanical Actuators, at the kinetic energies seen in Dragon 2 docking.

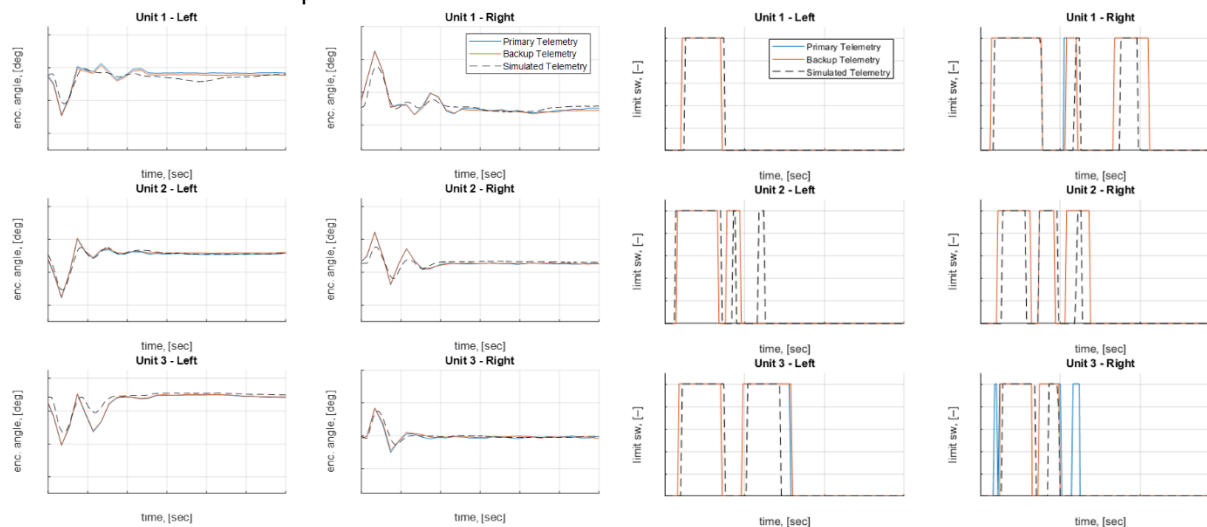
A critical finding from testing was that the capture switches could be triggered or cycled while the mechanism was captured, under heavy shear/bending load. This effect could be replicated with the simulation, after updates to the simulation using a more advanced capture switch model. The effect results from an interaction between the specified free play between the latches and strikers, in addition to the location of the capture switches, nominally on the edge of the radius of the passive soft capture system ring. This led to an increase in a flight software persistence value, specifically the number of consecutive

control cycles where the capture switch is not pressed to initiate the failed capture response, to account for this affect and prevent false negatives during the mission.

The high accuracy of the simulation meant that properly modeling the gravity off-loader became significant. To better replicate the on-orbit dynamics, the SCS ring has an off-loader which reacts the force of gravity from the SCS ring. However, this somewhat simple device can have significant frictions and stictions which become difficult to model for each unique run. A lesson learned from SDTS testing is to improve the repeatability of the gravity off-loader and fully characterize its behavior to aid in model correlation.

### Flight Reconstruction

The Demo 1 test flight mission was reconstructed with the docking simulation and the flight telemetry. The simulation had good correlation to the flight contact and attenuation dynamics, as well as the retraction and hard capture dynamics. Figure 8 compares the Flight Telemetry of the Attenuator Arm Encoders to the simulated encoders for capture.

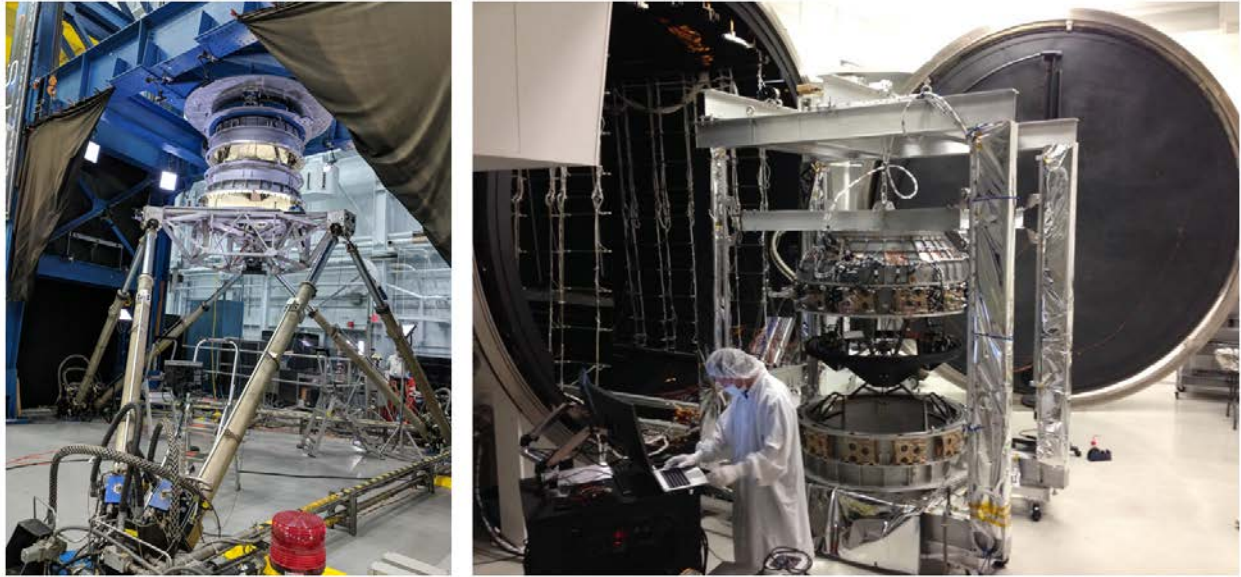


*Figure 8: Flight Attenuator Arm Telemetry vs. Simulation vs. Time*

For undocking, there were two principal sources used for correlation, the Ready to Hook switches, and the calculated Dragon velocity. The ready to hook switch states were used to verify the hook dynamics and overall Dragon separation state. The calculated Dragon velocity was used to verify the forces separating the Dragon from the ISS. Notably, there appeared to be little to no seal stiction (expected given the seal lubrication), and the latches provided nearly no restraining force, resulting in significant departure velocity when the hard capture hooks cleared. Properly modeling the hooks and their motion, in order to determine when there is clearance for departure, is a critical lesson to modeling undocking.

### **Qualification Testing Failures and Lessons Learned**

There were many lessons learned across the various test campaigns, several of which manifested as test failures. These early testing lessons learned led to design updates and delta qualification testing before the successful Demo 1 flight test and ultimately a more reliable system. A subset of these failures and their corrective actions are outlined below.



*Figure 9: (Left): The SpaceX undergoes dynamic testing on the NASA JSC Six-Degree-of-Freedom Dynamic Test System. (Right): The SpaceX Docking System prepared to enter thermal vacuum testing to perform system level qualification tests.*

#### TVAC Harnessing Failure

During system level thermal vacuum testing, there was a loss of signal from the controllers during the soft capture deploy sequence. To troubleshoot, the vacuum chamber was opened, and it was determined that the connectors to the soft capture latch controllers had sheared. Subscale testing was performed, and it was determined that the overwrap material of the harness, a rubber-like material, was stiffening the harness in vacuum, which caused it to shear. By replacing this overwrap material with a more flexible material, this condition was eliminated. This provides a valuable lesson learned in harnessing material selection, especially in dynamic mechanisms operating at a wide temperature range.



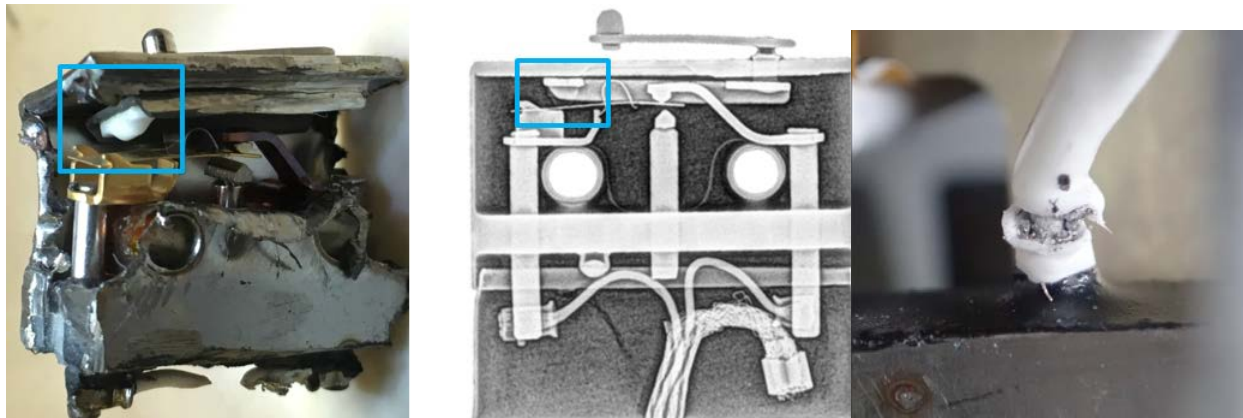
*Figure 10: Harness Connector Sheared during Deployment*



#### Microswitch Testing Defects

Microswitches are used throughout the SpaceX docking system as position indicators. In addition to the soft capture petal switches, which are used to declare capture, and the ready to hook switches, which indicate when the IDA hard capture plane has been contacted, switches are used within the attenuator arms and the hard capture hooks to indicate the mechanism state.

During system and sub-scale testing, microswitch failures were observed. Investigation revealed fracture of internal components to the microswitch caused during shock events. These failures lead to the selection of a different series of microswitches with fully potted internals that were more robust to shock events. This was an important lesson learned in assessing the robustness of commercial off the shelf components to shock induced failures. Additionally, it was found that the harness stiffness was such that it was subject to fatigue failure modes, even from the most careful of handling. A less stiff harness material was used to prevent this failure mode. This was an important lesson in selecting harness potting materials that are robust to nominal handling.



*Figure 11: Fracture of Ceramic Contact Pin in Shock (left) and undamaged switch (center); harness damage (right)*

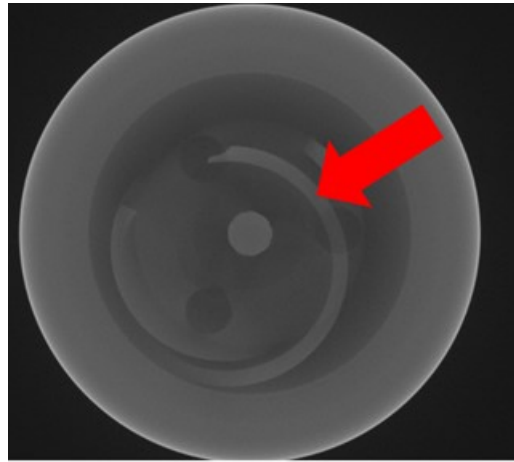
#### **Inspection Lessons Learned**

In addition to thoroughly testing hardware, the SpaceX docking system has gone through intensive inspection campaigns. Visual inspections and tear downs are a critical component of qualification campaign. SpaceX has also greatly increased its use of Computerized Tomography (CT) inspections. This allows hardware internals to be inspected without a tear down that inherently changes the condition. CT inspection of hardware has proven to be an immense resource for both failure investigations as well as identifying potential failure modes. Two examples of SpaceX's use of CT scanning are outlined. It is important to note that CT scans do provide radiation dosage to active components. A dosimeter scanned with a development unit can estimate this dosage to determine impact on the unit's usable life.

#### Retaining Ring Vendor Verification

After qualification testing, the eddy current damper was CT scanned, though it performed nominally during the campaign. The scan revealed a significant defect: a retaining ring had become dislodged from its groove. If settled in a slightly different position, this retaining ring could become stuck in the gearbox, causing the damper to jam. This led to additional development testing, where it was discovered that only improperly seated retaining rings were at risk of becoming dislodged; properly seated retaining rings survived qualification environments as indicated by pre-and-post environment CT scanning. Through this discovery, SpaceX worked with the vendor to implement robust quality control practices on retaining ring install. CT scan videos are stored for future evaluation, which proves useful if a different failure mode was discovered: videos can be reviewed without requiring hardware to be removed and re-scanned. This was an example where CT scanning enabled discovery of a potential problem without experiencing a dramatic

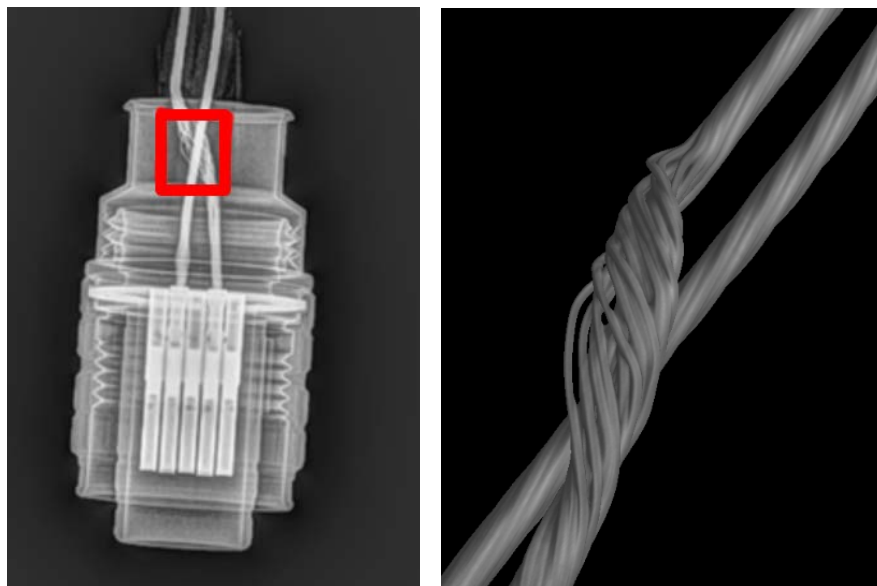
failure; CT scanning of both vendor and internal assembled components can be utilized as a very informative and robust quality control tool.



*Figure 12: Dislodged Retaining Ring*

#### Harness Failure

During an acceptance test, an intermittent connection was noted on one of the power and data buses. The problem was isolated to one connection. The controller was removed and tested and no anomalies noted, which led to additional investigation of the harnessing. It was found that 'wiggling' the harness could cause the bus to disconnect. The harness was CT scanned to identify the root cause: improper insulation on one wire allowed the cabling to 'bird cage', which could then short against the connector. The insulation was not properly adhering to the cable, which would allow the insulation to pull off the cable and induce this condition. CT scanning provides extremely clear indications of this root cause. This issue could have been easily written off as a single workmanship issue; however, through CT scan it was identified that this was not an isolated incident but cable adhesion susceptibility; and through further testing the faulty cable lots were identified and contained. This is another example where CT scanning can be a valuable investigation tool when failures occur.



*Figure 13: CT Scan images showing 'bird caging' of cable which had its insulation improperly adhered and pulled up. This allowed the cable to short against the connector housing*

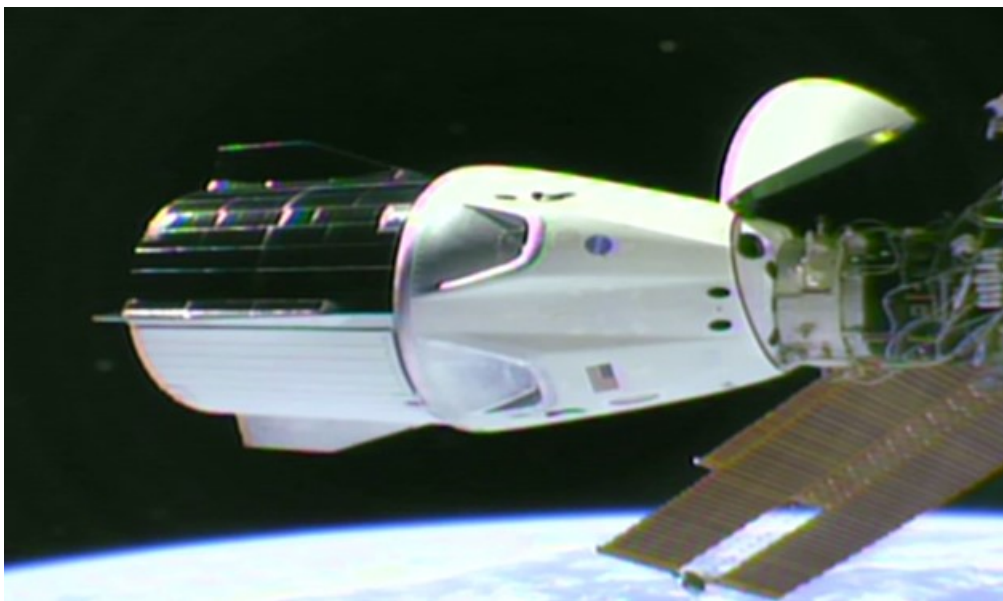
### **Lessons Learned: Systems Interactions**

SpaceX benefits from having a highly integrated team of software, avionics, operations, GNC and mechanisms engineers. This integration is critical to the successful design and continued operation of the SpaceX docking system. The close interactions between flight software and the mechanical team, for example, allowed for very robust testing of both hardware and software: the flight software algorithms, fault detection methods, voting schemes and responses were used for all system level testing. This allowed early validation that the software responded as expected, including in off-nominal cases. This close interaction also allowed validation of configurable parameters, and the accurate modeling of flight software response in simulation allowed small changes in these parameters to be validated. For example, a key parameter in the docking configuration is the persistence value to declare soft capture and loss of capture from the soft capture switches. By using flight software during 6-DoF testing at JSC, and then adjusting this configurable value in the docking simulation, this parameter could be fine-tuned for the desired response. The close interaction between the simulation and GNC allowed rapid iteration based on expected contact conditions and docking parameters.

The integration between avionics teams and hardware teams was also critical to the rapid and successful development of the SpaceX docking system. Controllers were developed specifically for the docking system application, and close coordination during subassembly and system level testing allowed for the fine-tuning of controller firmware. With a dynamic system, harness routing to the soft capture latches and switches is critical; a misrouted harness could impede motion and have drastic impacts on mechanism success. Close coordination with harnessing engineers, including involving them in system level testing so they could better understand the expected range of motion, allowed for a successful harness design.

### **Conclusions**

The SpaceX docking system is a custom implementation of the NASA Docking System architecture. It is the first American designed and manufactured docking system to be flight proven in decades. The SpaceX docking system leverages a semi-passive system to effect docking capture, attenuation, and alignment before retraction to hard capture. It is effective for a broad range of contact conditions, is low power, light weight, and cost effective. The system has been extensively simulated, tested, and inspected to ensure successful operation throughout a wide range of possible contact conditions.



*Figure 13: The Demo 1 Crew Dragon Capsule Docked to the ISS Forward IDA in March 2019.*

## References

1. McFatter, J., K. Keiser, and T. Rupp, "NASA Docking System Block 1: NASA's New Direct Electric Docking System Supporting ISS and Future Human Space Exploration" *Proceedings of the 44th Aerospace Mechanisms Symposium, NASA Glenn Research Center, May 16-18, 2018*
2. NASA Docking System Interface Definitions Document (IDD), JSC 65795, Revision J